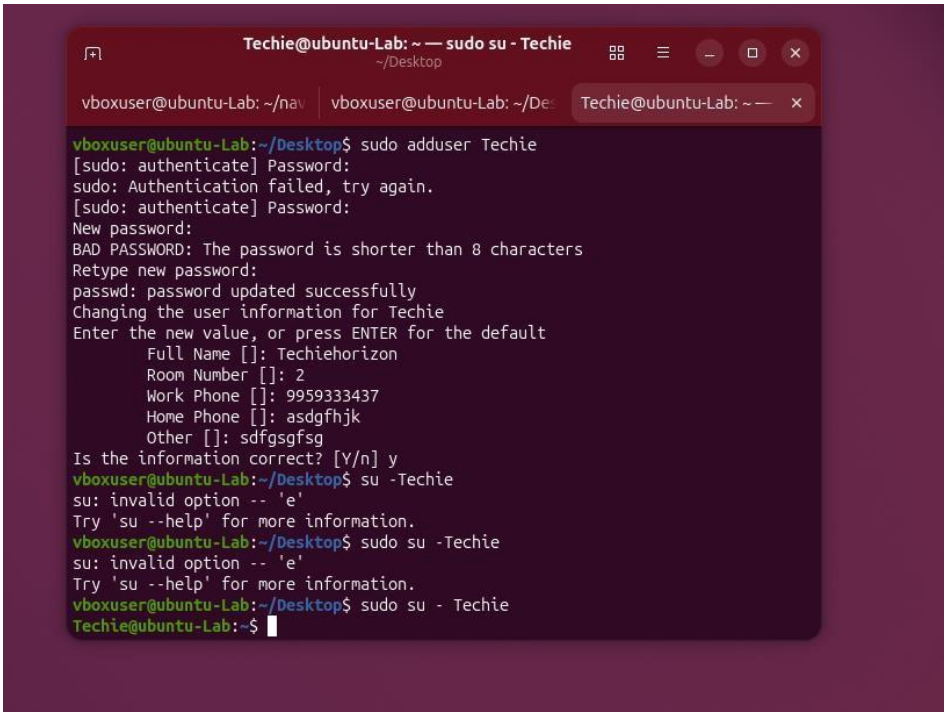
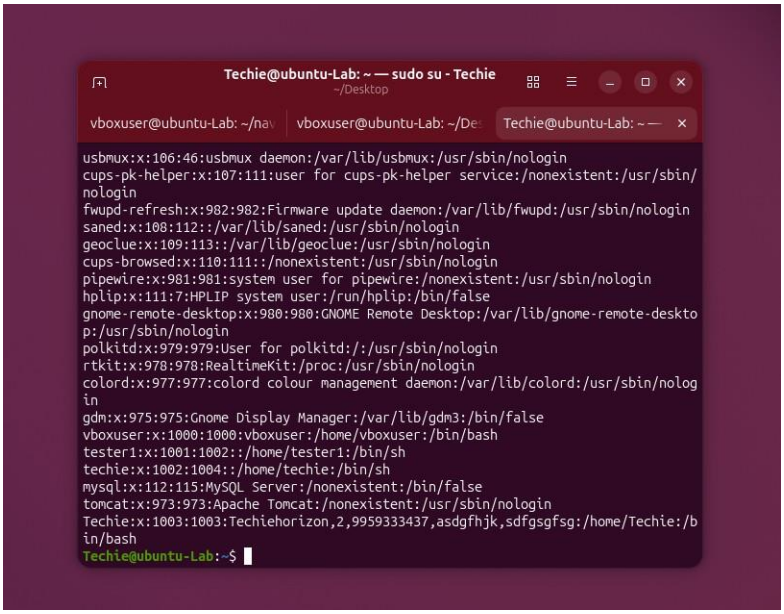
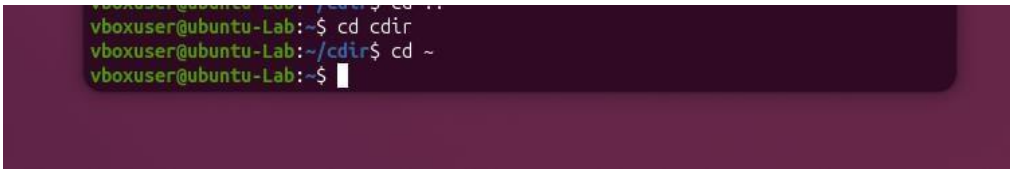
1. Create user with name Techie and provide sudo access to user.





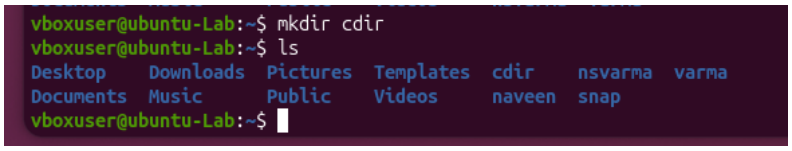
2. Navigate to the home directory.

Cd ~



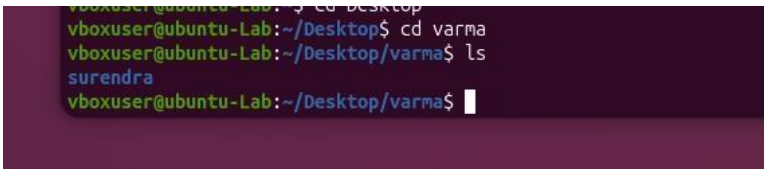
3. Create a new directory.

   Mkdir



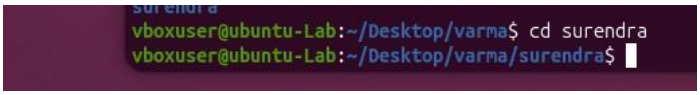
4. List the contents of a directory.

   Ls



5. Change the current directory.

   Cd directoryname



6. Create a new empty file.

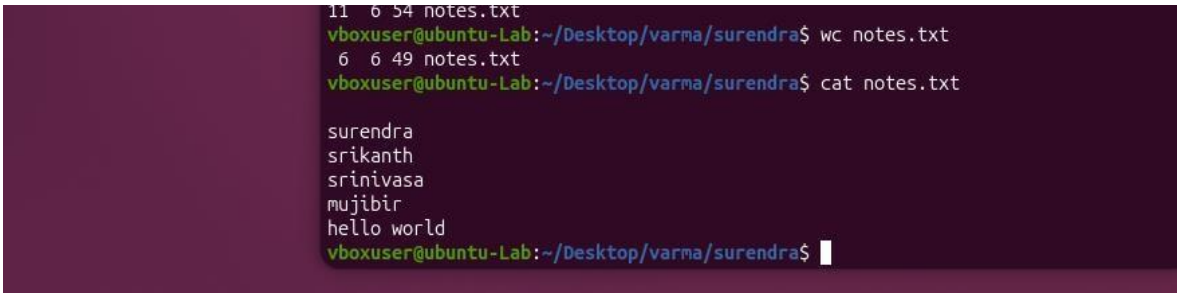
   Touch note.txt



7. View the contents of a file.

   Vi notes.txt wq!

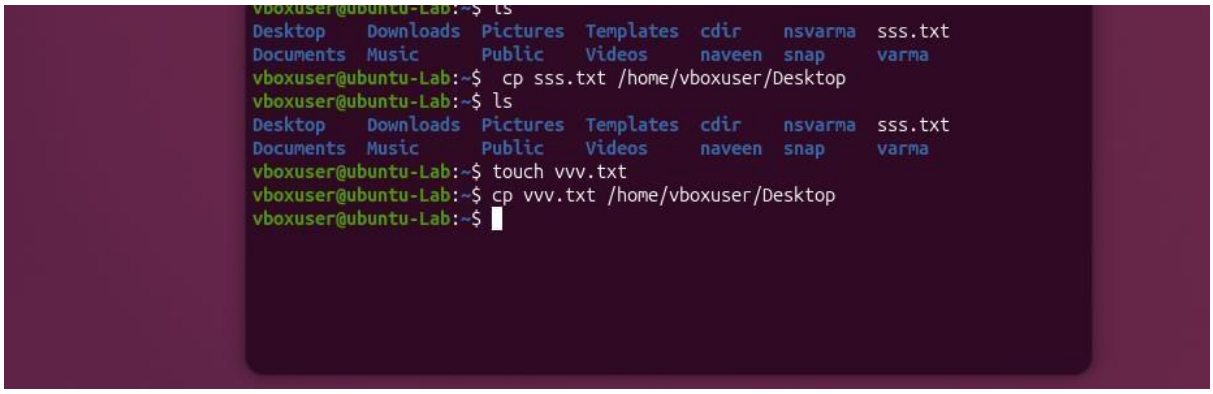
   Cat notes.txt



8. Copy a file to another location.

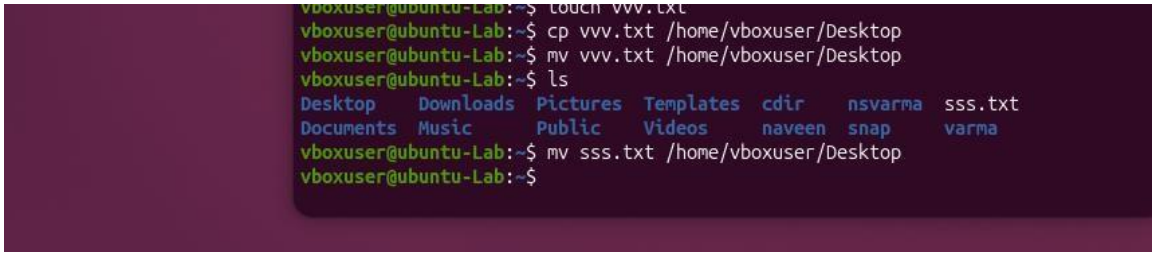
   Touch sss.txt (to create a file)

   Cp filename (Path)



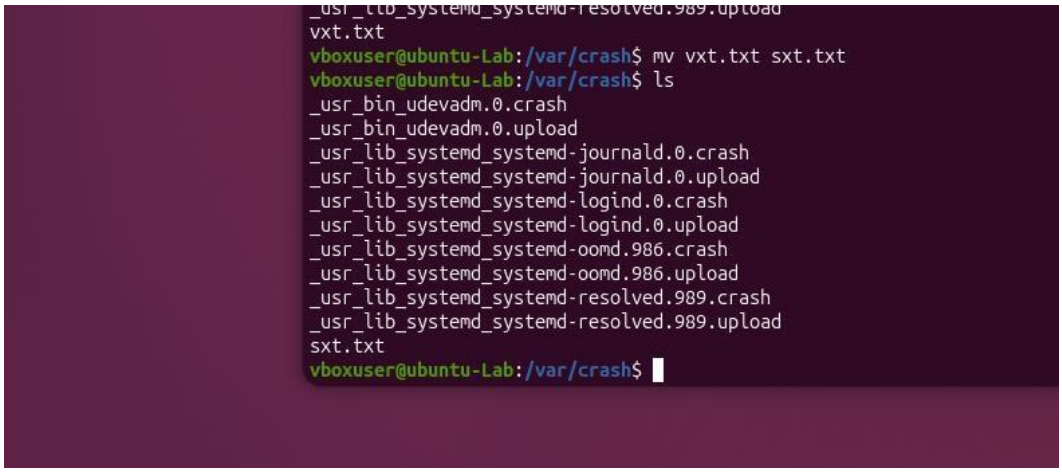
9. Move a file to another location.

Mv sss.txt (filename) path



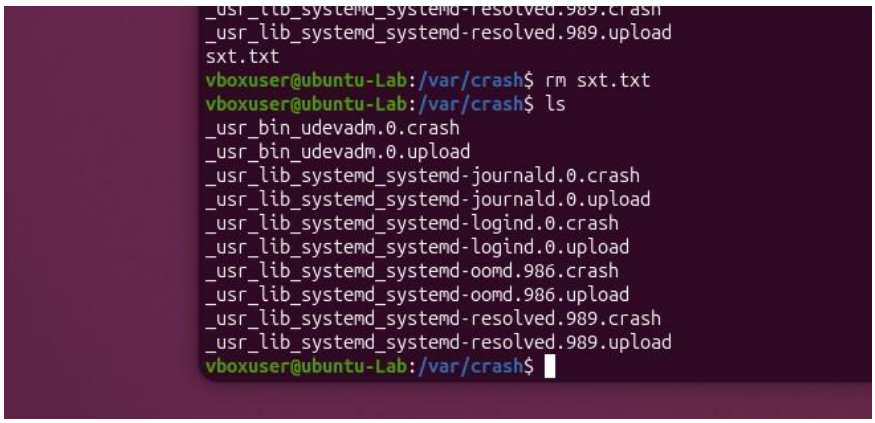
10. Rename a file.

Mv nxt.txt vxt.txt



11. Delete a file.

Rm vxt.txt



12. Grant or revoke permissions on a file or directory.

Chmod change permissions

Chown ownership

chgrp

rwx **owner** (user) permissions

r-x **group** permissions

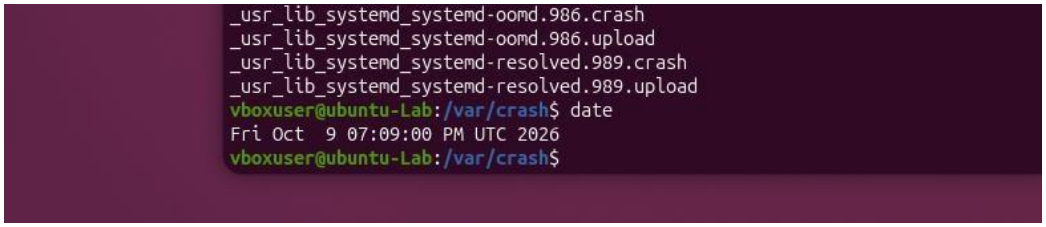
R-- others permissions

Read r 4

Write w 2
Execute x 1

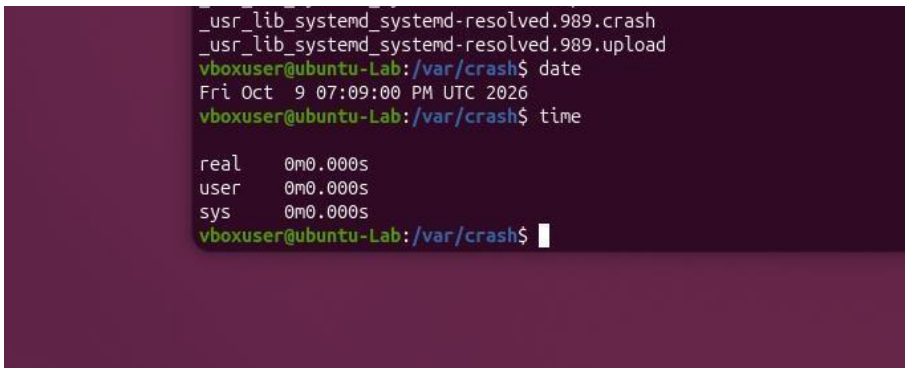
13. View the current date and time.

Date



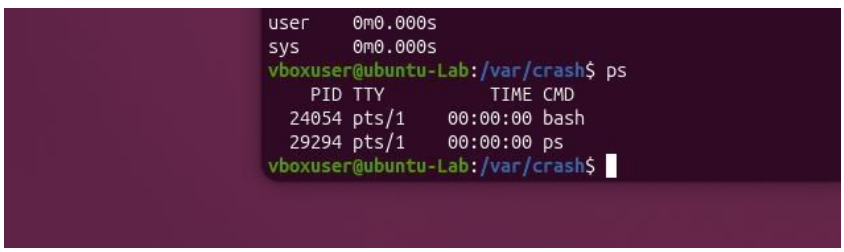
14. Check the system uptime.

Time



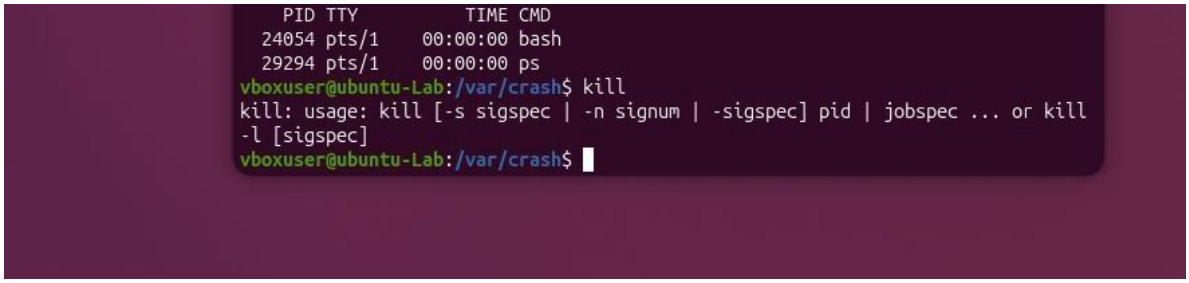
15. View the running processes.

Ps



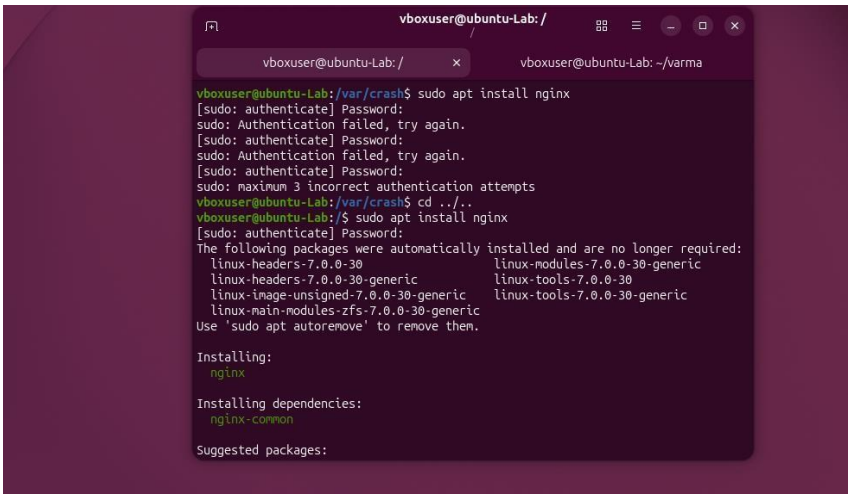
16. Kill a running process.

Kill



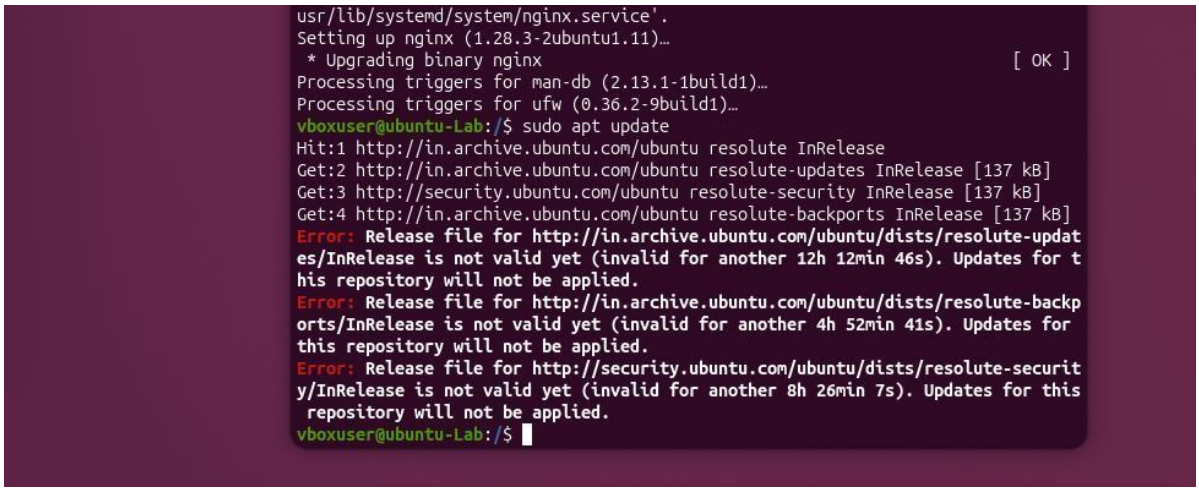
17. Install a package using the package manager (e.g., apt or yum).

Sudo apt install nginx



18. Update the system packages.

Sudo apt update



19. Create a symbolic link.

mkdir LinuxPractice

cd LinuxPractice

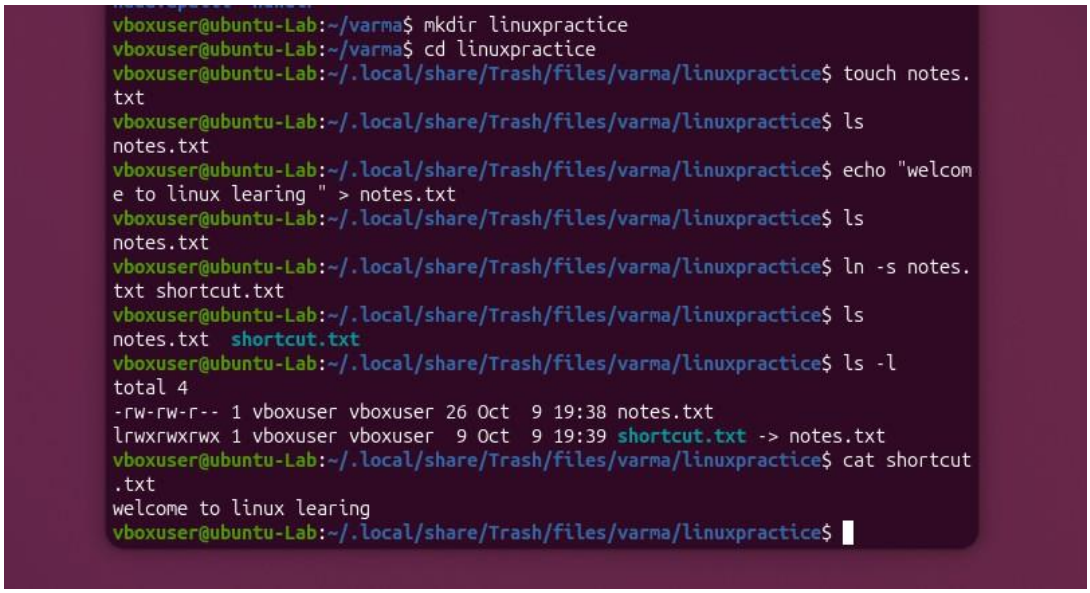
touch notes.txt

echo "Welcome to Linux learning" > notes.txt

ln -s notes.txt shortcut.txt

ls -l

cat shortcut.txt



20. Search for files using the find command.

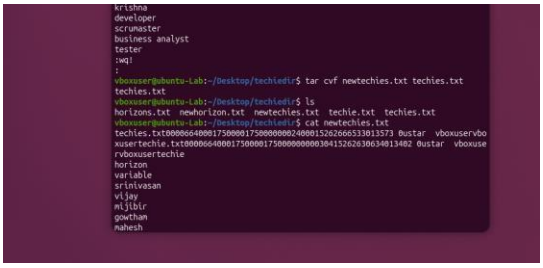
Find

```
vboxuser@ubuntu-Lab:~/.local/share/Trash/files/varma/linuxpractice$ cat shortcut
.txt
welcome to linux learing
vboxuser@ubuntu-Lab:~/.local/share/Trash/files/varma/linuxpractice$ find
.
./shortcut.txt
./notes.txt
vboxuser@ubuntu-Lab:~/.local/share/Trash/files/varma/linuxpractice$
```

21. Compress and decompress files using tar.

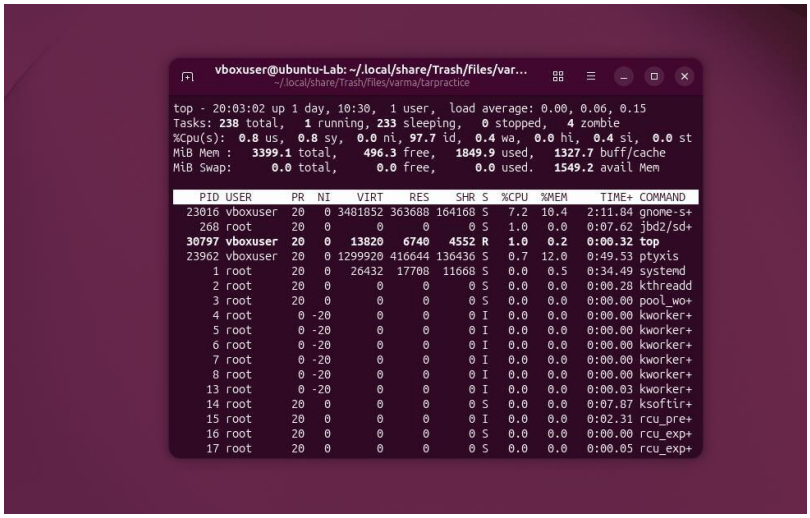
Tar cvf newfilename oldfilename

Tar xuf newfilename

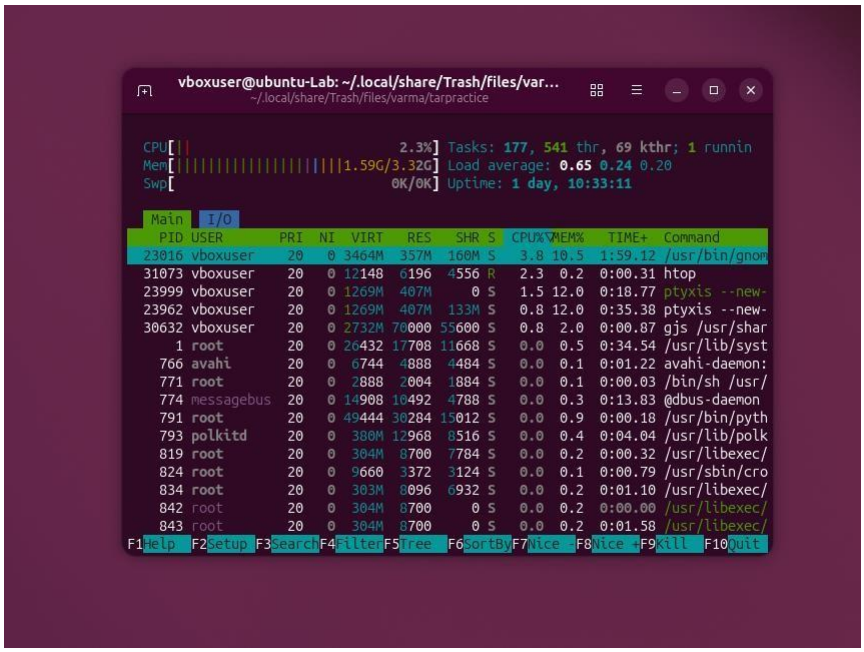


22. Monitor system resources with top or htop.

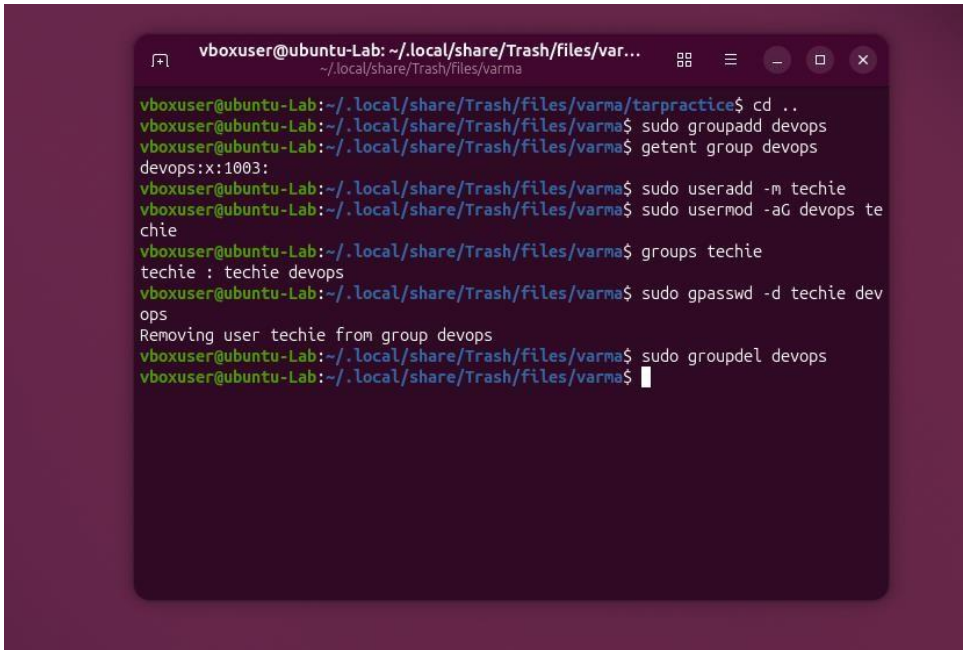
Top



Htop



23. Create and manage user groups.



24. Set up SSH password less authentication.

```
Created symlink '/etc/systemd/system/sshd@.service.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/ssh.socket.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Processing triggers for man-db (2.13.1-1build1)…
Processing triggers for ufw (0.36.2-9build1)…
vboxuser@Linux:~/devops$ sudo systemct1 enable --now ssh
sudo: 'systemct1': command not found
vboxuser@Linux:~/devops$ sudo systeml enable --now ssh
sudo: 'systeml': command not found
vboxuser@Linux:~/devops$ sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/sshd.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
vboxuser@Linux:~/devops$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-10-10 12:04:33 IST; 51s ago
 Invocation: 983611cb1e7b49789b90e86fc19eccc5
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 18750 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 18752 (sshd)
      Tasks: 1 (limit: 9411)
     Memory: 1.2M (peak: 2.2M)
        CPU: 14ms
     CGroup: /system.slice/ssh.service
             └─18752 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 10 12:04:33 Linux systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 10 12:04:33 Linux sshd[18752]: Server listening on 0.0.0.0 port 22.
Oct 10 12:04:33 Linux sshd[18752]: Server listening on :: port 22.
Oct 10 12:04:33 Linux systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
vboxuser@Linux:~/devops$
```

```
vboxuser@Linux: ~ — ssh -o PreferredAuthentications=publickey -o PasswordAuthentication=no vboxuser@localhost
~/Downloads
vboxuser@Linux:~/Downloads$ ssh -o PreferredAutthentications=publickeylocalhost
command-line: line 0: Bad configuration option: preferredautthentications
vboxuser@Linux:~/Downloads$ chmod 600 ~/.ssh/authorized_keys
vboxuser@Linux:~/Downloads$ ssh -o PreferredAuthentications=publickey -o PasswordAuthentication=novboxuser@localhost
command-line line 0: unsupported option "novboxuser@localhost".
vboxuser@Linux:~/Downloads$ ^C
vboxuser@Linux:~/Downloads$ ssh -o PreferredAuthentications=publickey -o PasswordAuthentication=no vboxuser@localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is: SHA256:8YYN2+aL4BeYh/zBVVki0fsjhztblWXL172AYqpH264
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

vboxuser@Linux:~$
```

```
vboxuser@Linux: ~ — ssh -v -o PreferredAuthentications=publickey -o PasswordAuthentication=no localhost
~/Downloads
debug1: Will attempt key: /home/vboxuser/.ssh/id_ed25519_sk
debug1: Offering public key: /home/vboxuser/.ssh/id_ed25519 ED25519 SHA256:dkhhMOtIzxHdsYE4QxIHACALdy4m9fMkRmwn3oCW2+w agent
debug1: Server accepts key: /home/vboxuser/.ssh/id_ed25519 ED25519 SHA256:dkhhMOtIzxHdsYE4QxIHACALdy4m9fMkRmwn3oCW2+w agent
Authenticated to localhost ([127.0.0.1]:22) using "publickey".
debug1: channel 0: new session [client-session] (inactive timeout: 0)
debug1: Requesting no-more-sessions@openssh.com
debug1: Entering interactive session.
debug1: pledge: filesystem
debug1: client_input_global_request: rtype hostkeys-00@openssh.com want_reply 0
debug1: client_input_hostkeys: searching /home/vboxuser/.ssh/known_hosts for localhost / (none)
debug1: client_input_hostkeys: searching /home/vboxuser/.ssh/known_hosts2 for localhost / (none)
debug1: client_input_hostkeys: hostkeys file /home/vboxuser/.ssh/known_hosts2 does not exist
debug1: client_input_hostkeys: no new or deprecated keys from server
debug1: Remote: /home/vboxuser/.ssh/authorized_keys:1: key options: agent-forwarding port-forwarding pty user-rc x11-forwarding
debug1: Remote: /home/vboxuser/.ssh/authorized_keys:1: key options: agent-forwarding port-forwarding pty user-rc x11-forwarding
debug1: Sending environment.
debug1: channel 0: setting env COLORTERM = "truecolor"
debug1: channel 0: setting env LANG = "en_US.UTF-8"
debug1: pledge: fork
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

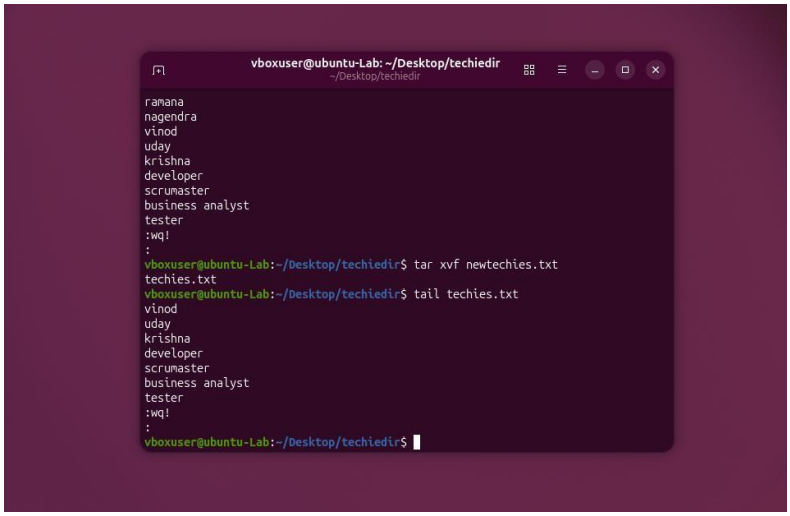
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

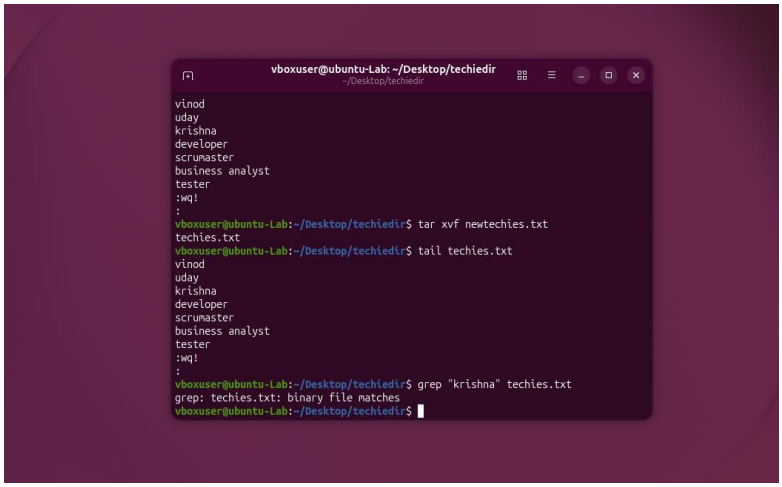
Last login: Sun Oct 11 13:15:31 2026 from 127.0.0.1
vboxuser@Linux:~$
```

25. Monitor log files using tail or grep.

Tail – last 10 lines in the file

Grep – particual text



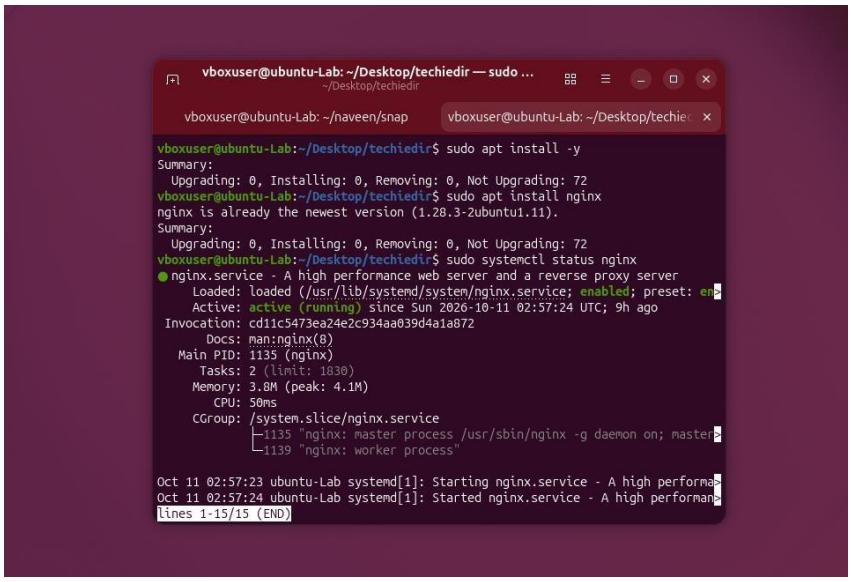


26. Set up a web server (e.g., Apache or Nginx).

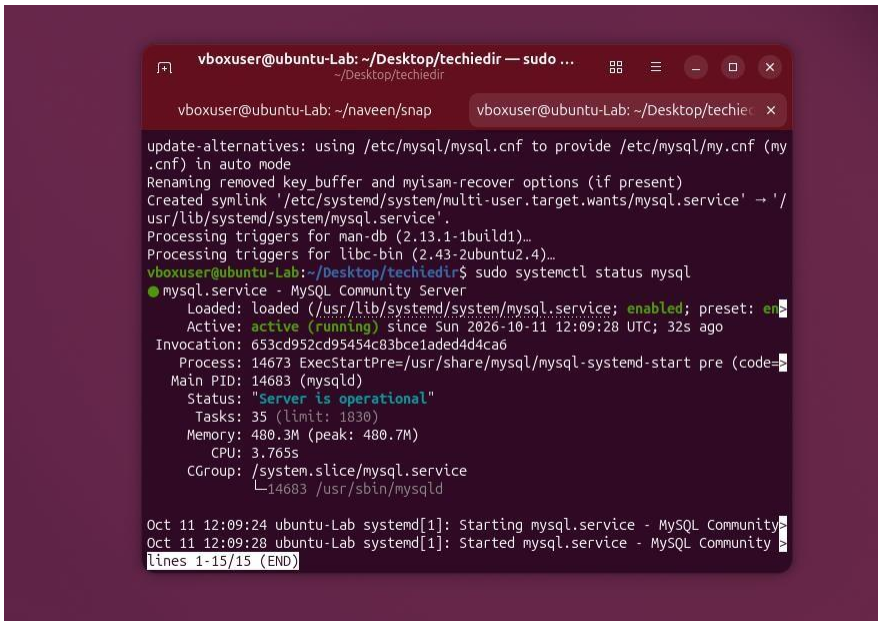
Sudo apt update

Sudo apt install ngnix

Sudo systemctl status nginx



27. Configure and secure a MySQL Database.

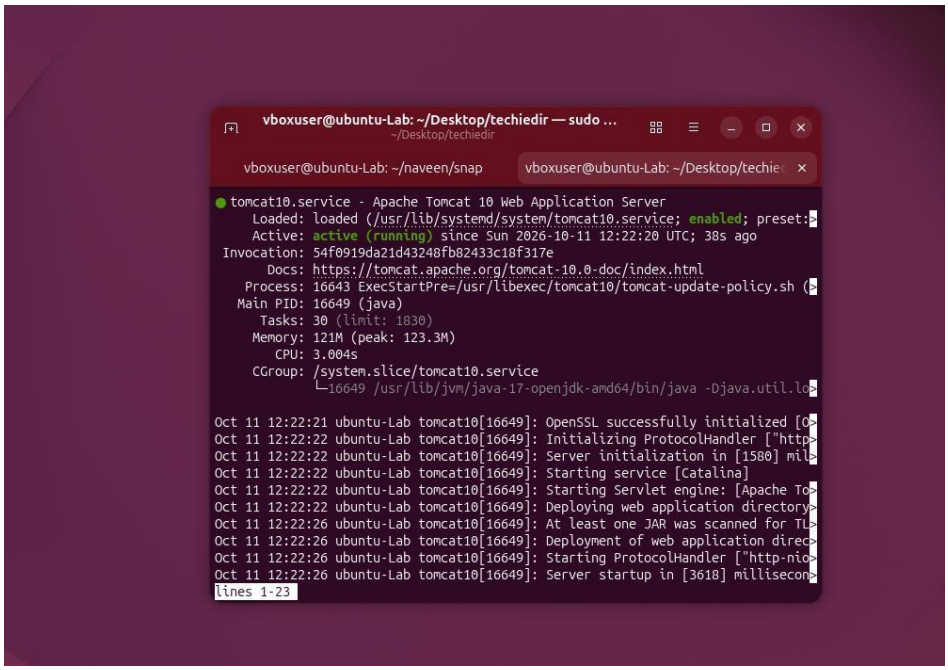


sudo apt update

sudo apt install mysql-server

sudo systemctl status mysql

28. Set up a Application Server (e.g.,Apache Tomcat)



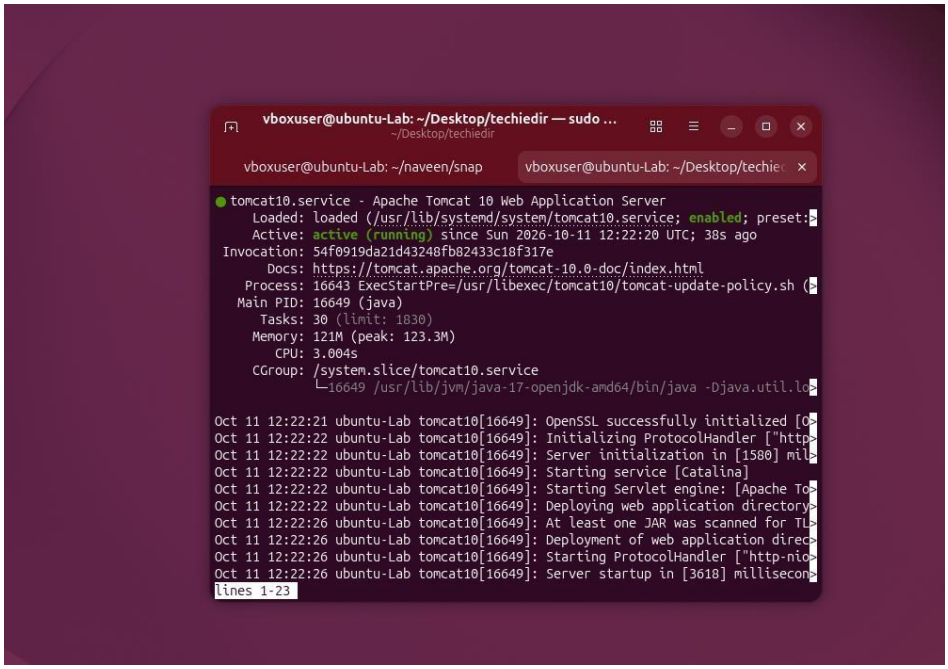
sudo apt update

sudo apt install openjdk-17-jdk tomcat10 -y

sudo systemctl enable --now tomcat10

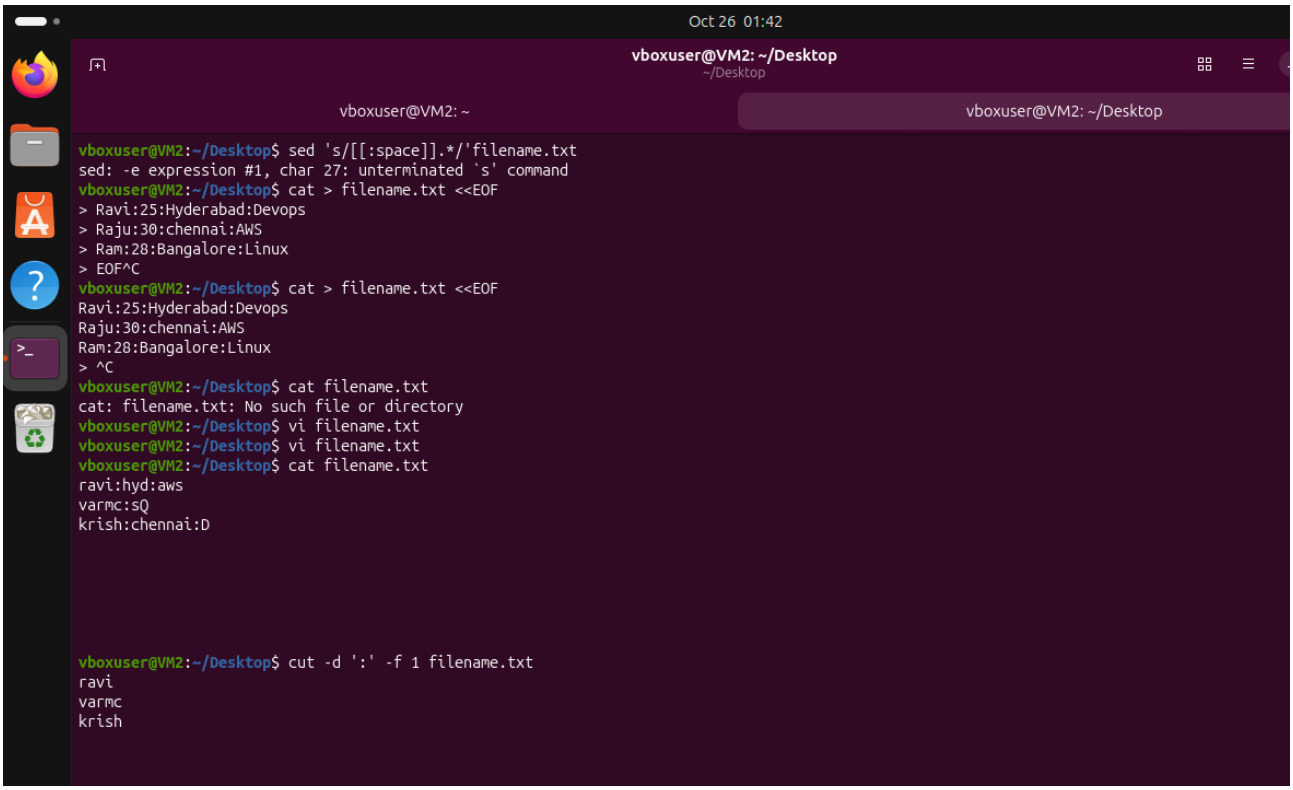
sudo systemctl status tomcat10

29. create a service file for Apache Tomcat.(Should execute by using systemtctl command)

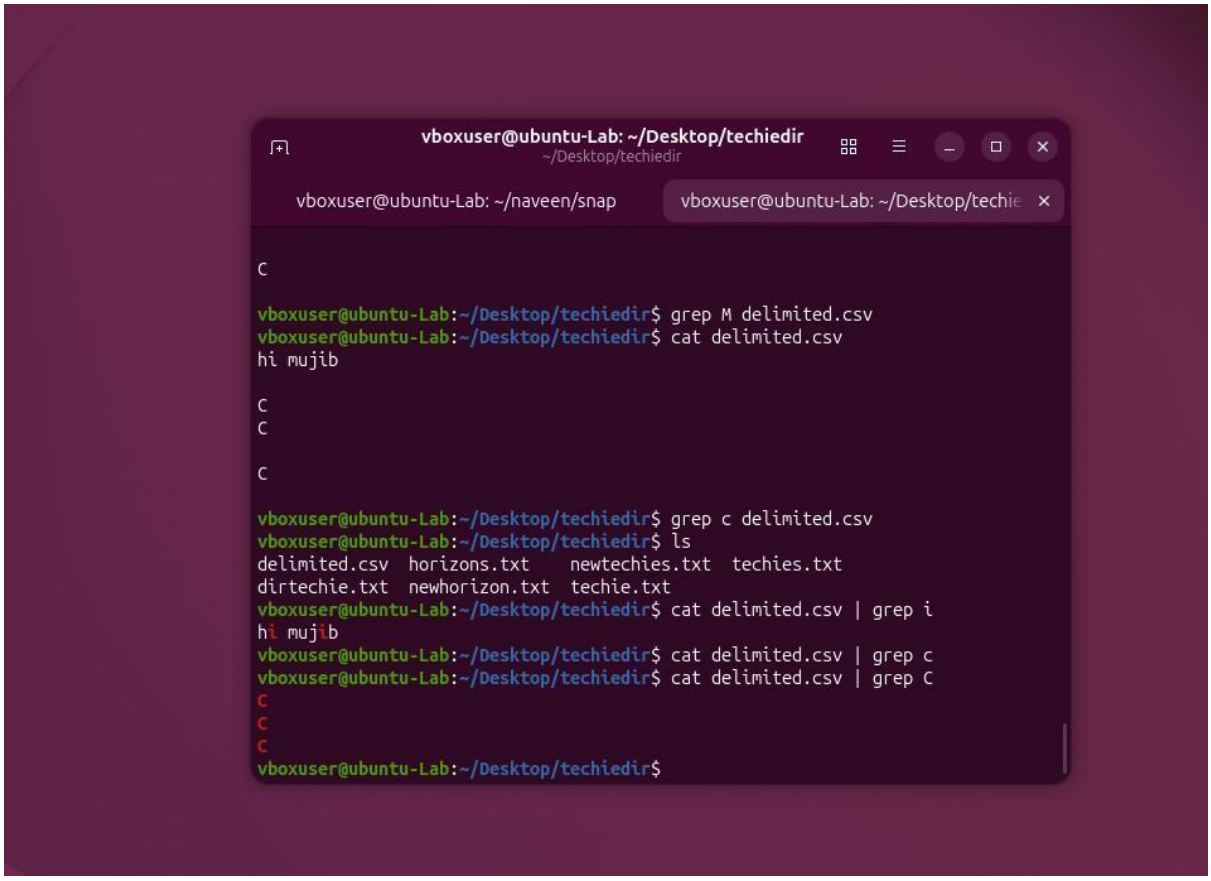


30. Print specific columns from a delimited file.

cut -d ':' -f 1 filename.txt



31. Filter and print lines based on a specific pattern or condition.



Cat delimited.csv | grep C

32. Calculate and print the average, sum, or other statistics of a column.

```
 Create a sample file
Vi filenum.txt
10
20
30
40
50
EOF

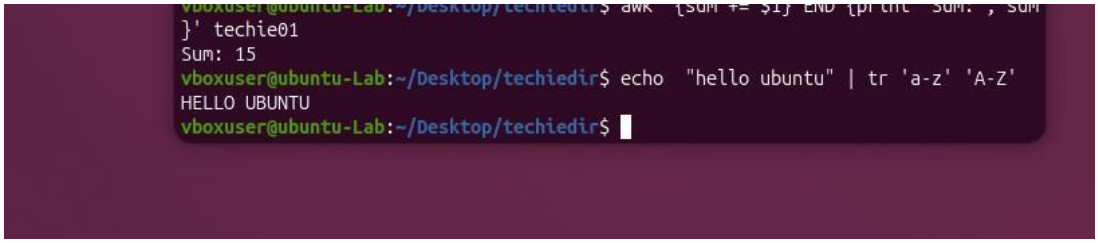
paste -sd+ numbers.txt | bc
150

echo "scale=2; $(paste -sd+ numbers.txt | bc) / $(wc -l < numbers.txt)" | bc
30.00

sort -n numbers.txt | tail -1
50
sort -n numbers.txt | head -1
10
wc -l < numbers.txt
5
```

- cut — extract a column.
- paste — combine values into one line.
- bc — perform arithmetic calculations.
- sort -n — sort numbers.
- wc -l — count lines.

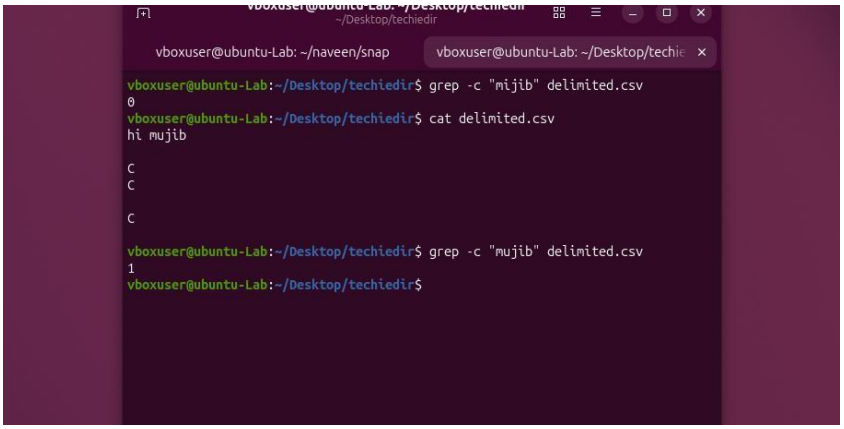
33. Perform string manipulation, such as extracting substrings or changing case.



echo "hello ubuntu" | tr 'a-z' 'A-Z'

echo "HELLO UBUNTU" | tr 'A-Z' 'a-z'

34. Count the occurrences of a specific pattern in a file.



Grep -c "mujib" filename.txt

35. Sort lines based on a specific field or column.



36. Merge multiple files based on a common field or column.



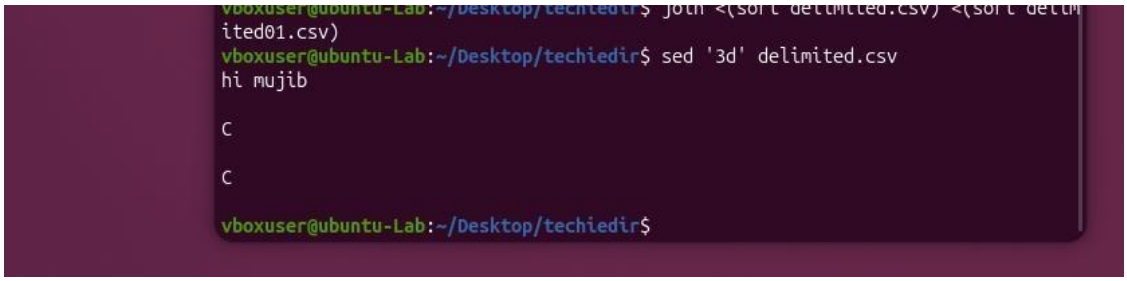
join <(sort file1.txt) <(sort file2.txt)

37. Substitute text in a file using search and replace.



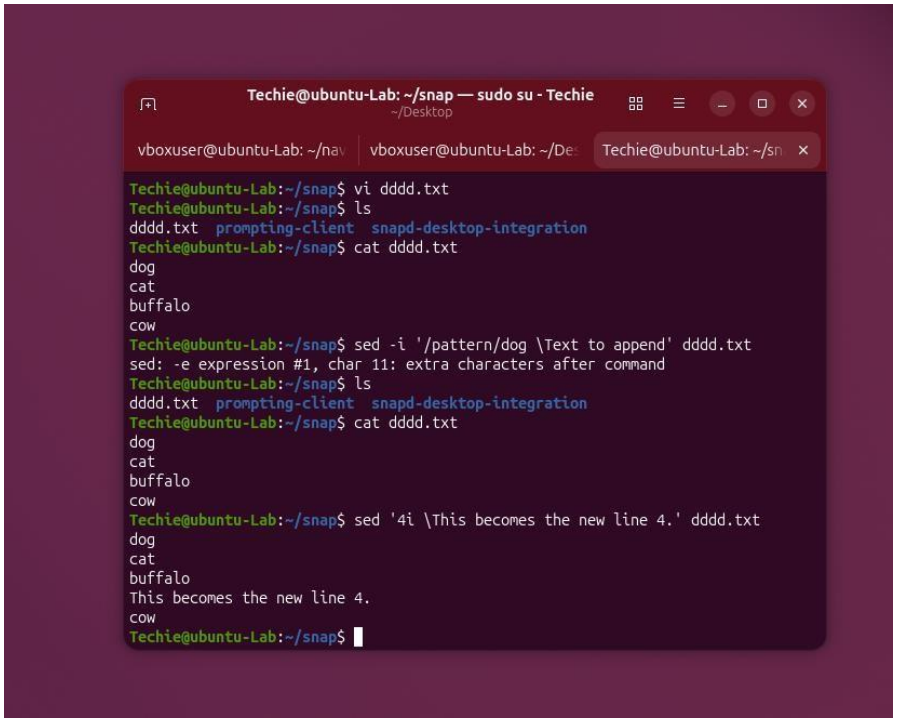
Sed ‘s/mujib/MUJI/’ delimited.csv

38. Delete specific lines based on a pattern or line number.

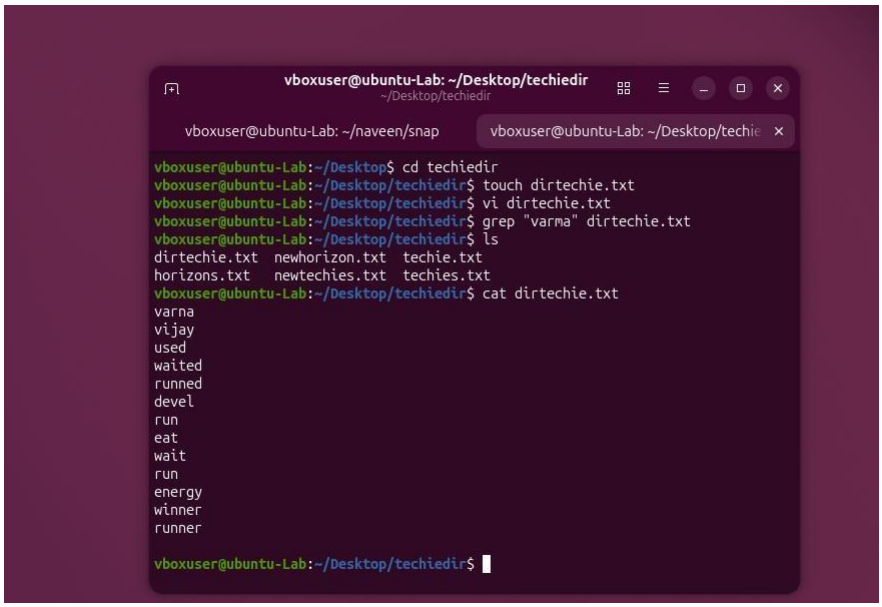


sed '5d' filename.txt

39. Append or insert text before or after a specific pattern or line.



40. Print only specific lines from a file.

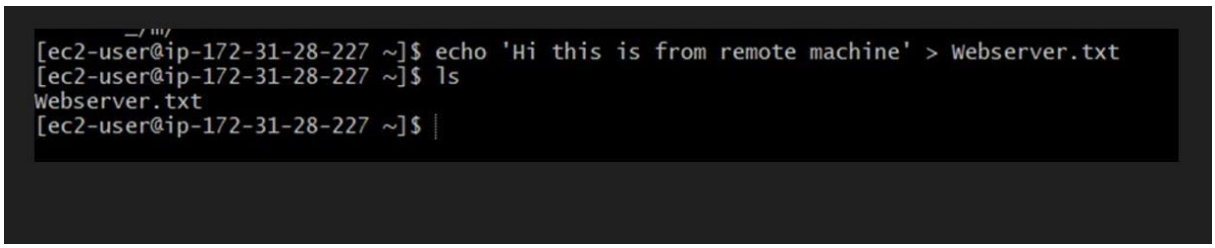


Grep “varna” dirtechie.txt

sed -n '5p' file.txt

sed -n '5,10p' file.txt

41. Copy file from linux to windows machine





42. 5 use cases for AWK and 5 use cases for sed.

```
vboxuser@Linux:~/devops$ cat employees.txt
101 Ravi Devoloper 25000
102 Priya Tester 30000
104 Arun Manager 45000
104 Sita Developer 35000
105 Rahul Tester 28000
vboxuser@Linux:~/devops$ awk '{print $2}' employees.txt
Ravi
Priya
Arun
Sita
Rahul
vboxuser@Linux:~/devops$ awk '{print $3}' employees.txt
Devoloper
Tester
Manager
Developer
Tester
vboxuser@Linux:~/devops$ awk '{print $4}' employees.txt
25000
30000
45000
35000
28000
vboxuser@Linux:~/devops$ awk '$4 > 30000' employees.txt
104 Arun Manager 45000
104 Sita Developer 35000
vboxuser@Linux:~/devops$ awk '{sum += $4} END {print "Total salary:", sum}' employees.txt
Total salary: 163000
vboxuser@Linux:~/devops$
```

```
vboxuser@Linux:~/devops$ awk '{sum += $4} END {print "Total salary:", sum}' employees.txt
Total salary: 163000
vboxuser@Linux:~/devops$ sed -n '1,3p' employees.txt
101 Ravi Devoloper 25000
102 Priya Tester 30000
104 Arun Manager 45000
vboxuser@Linux:~/devops$ sed 's/Developer/Engineer/g' employees.txt
101 Ravi Devoloper 25000
102 Priya Tester 30000
104 Arun Manager 45000
104 Sita Engineer 35000
105 Rahul Tester 28000
vboxuser@Linux:~/devops$ sed '/Tester/d' employees.txt
101 Ravi Devoloper 25000
104 Arun Manager 45000
104 Sita Developer 35000
vboxuser@Linux:~/devops$ sed 's/Ravi/Ravi Kumar/' employees.txt
101 Ravi Kumar Devoloper 25000
102 Priya Tester 30000
104 Arun Manager 45000
104 Sita Developer 35000
105 Rahul Tester 28000
vboxuser@Linux:~/devops$ sed -n '/Manager/p' employees.txt
104 Arun Manager 45000
```